

# Data Dictionary — Vidya Setu Foundation Impact Dashboard

## Overview

This document describes all data fields used in the Impact Measurement Dashboard for Vidya Setu Foundation. The data covers **500 beneficiaries**, **1,200 programme activities**, and **18 months** of operations across **10 districts** in 5 Indian states.

## 1. NGO Profile ( ngo\_profile.json )

| Field               | Type          | Description              | Example                        |
|---------------------|---------------|--------------------------|--------------------------------|
| name                | String        | Name of the NGO          | "Vidya Setu Foundation"        |
| mission             | String        | Mission statement        | "Bridging the learning gap..." |
| founded             | Integer       | Year of establishment    | 2018                           |
| headquarters        | String        | HQ location              | "Pune, Maharashtra"            |
| operating_districts | Array[String] | List of active districts | ["Dharwad", "Raichur", ...]    |
| operating_states    | Array[String] | List of active states    | ["Karnataka", "UP", ...]       |
| total_staff         | Integer       | Total staff count        | 45                             |
| field_workers       | Integer       | Number of field workers  | 10                             |

| Field                          | Type          | Description                   | Example                    |
|--------------------------------|---------------|-------------------------------|----------------------------|
| <code>annual_budget_inr</code> | Integer       | Annual budget in INR          | 8500000                    |
| <code>programmes</code>        | Array[Object] | List of programme definitions | See Programmes table       |
| <code>funding_sources</code>   | Array[String] | Sources of funding            | ["CSR - Tata Trusts", ...] |

## 2. Beneficiaries ( `beneficiaries.json` )

Each record represents one beneficiary enrolled in Vidya Setu programmes.

| Field                        | Type          | Description           | Example                                              |
|------------------------------|---------------|-----------------------|------------------------------------------------------|
| <code>id</code>              | String        | Unique beneficiary ID | "BEN-0001"                                           |
| <code>name</code>            | String        | Anonymised name       | "Beneficiary_1"                                      |
| <code>gender</code>          | String        | Gender identity       | "Female" / "Male" / "Other"                          |
| <code>age_range</code>       | String        | Age bracket           | "6-10", "11-14", "15-18",<br>"19-25", "26-40", "41+" |
| <code>category</code>        | String        | Social category       | "General", "OBC", "SC", "ST"                         |
| <code>district</code>        | String        | Operating district    | "Dharwad"                                            |
| <code>state</code>           | String        | State                 | "Karnataka"                                          |
| <code>region</code>          | String        | Geographic region     | "South", "North", "West",<br>"East", "Central"       |
| <code>enrollment_date</code> | String (Date) | Date of enrolment     | "2025-03-15"                                         |

| Field                            | Type            | Description                             | Example          |
|----------------------------------|-----------------|-----------------------------------------|------------------|
| <code>enrolled_programmes</code> | Array[String]   | Programme IDs enrolled in               | ["P001", "P002"] |
| <code>baseline_reading</code>    | Integer (0-100) | Pre-intervention reading score          | 35               |
| <code>baseline_math</code>       | Integer (0-100) | Pre-intervention math score             | 30               |
| <code>baseline_digital</code>    | Integer (0-100) | Pre-intervention digital literacy score | 15               |
| <code>current_reading</code>     | Integer (0-100) | Latest reading score                    | 48               |
| <code>current_math</code>        | Integer (0-100) | Latest math score                       | 42               |
| <code>current_digital</code>     | Integer (0-100) | Latest digital literacy score           | 35               |
| <code>reading_improvement</code> | Integer         | Change in reading score                 | +13              |
| <code>math_improvement</code>    | Integer         | Change in math score                    | +12              |
| <code>digital_improvement</code> | Integer         | Change in digital literacy score        | +20              |
| <code>attendance_rate</code>     | Float (0-1)     | Proportion of sessions attended         | 0.78             |
| <code>is_active</code>           | Boolean         | Currently active in programme           | true             |

### 3. Activities ( activities.json )

Each record represents one programme session/activity conducted in the field.

| Field            | Type          | Description                        | Example                                         |
|------------------|---------------|------------------------------------|-------------------------------------------------|
| id               | String        | Unique activity ID                 | "ACT-0001"                                      |
| programme_id     | String        | Programme identifier               | "P001"                                          |
| programme_name   | String        | Human-readable programme name      | "After-School Tutoring"                         |
| programme_type   | String        | Programme category                 | "education", "digital", "training", "community" |
| date             | String (Date) | Date of activity                   | "2025-06-12"                                    |
| month            | String        | Year-month for aggregation         | "2025-06"                                       |
| district         | String        | District where conducted           | "Dharwad"                                       |
| state            | String        | State                              | "Karnataka"                                     |
| region           | String        | Region                             | "South"                                         |
| field_worker     | String        | Name of field worker               | "Anita Sharma"                                  |
| num_participants | Integer       | Number of participants             | 18                                              |
| participant_ids  | Array[String] | IDs of participating beneficiaries | ["BEN-0012", "BEN-0045"]                        |
| cost_inr         | Float         | Cost of activity in INR            | 425.50                                          |

| Field                       | Type          | Description                      | Example                                   |
|-----------------------------|---------------|----------------------------------|-------------------------------------------|
| <code>funding_source</code> | String        | Source of funding                | "CSR - Tata Trusts"                       |
| <code>duration_hours</code> | Float         | Duration in hours                | 2.5                                       |
| <code>quality_flags</code>  | Array[String] | Data quality issues detected     | ["missing_attendance", "late_submission"] |
| <code>verified</code>       | Boolean       | Whether record has been verified | true                                      |

## Quality Flag Values

| Flag                            | Meaning                                      |
|---------------------------------|----------------------------------------------|
| <code>missing_attendance</code> | Attendance sheet not submitted               |
| <code>duplicate_entry</code>    | Possible duplicate record detected           |
| <code>date_mismatch</code>      | Activity date doesn't match reporting period |
| <code>incomplete_form</code>    | Required fields missing in submission        |
| <code>late_submission</code>    | Record submitted after deadline              |

## 4. Programmes ( `programmes.json` )

| Field             | Type   | Description          | Example                 |
|-------------------|--------|----------------------|-------------------------|
| <code>id</code>   | String | Programme identifier | "P001"                  |
| <code>name</code> | String | Programme name       | "After-School Tutoring" |

| Field                | Type    | Description                         | Example     |
|----------------------|---------|-------------------------------------|-------------|
| type                 | String  | Programme category                  | "education" |
| cost_per_session     | Integer | Baseline cost per session (INR)     | 350         |
| target_beneficiaries | Integer | Target number of beneficiaries      | 200         |
| target_sessions      | Integer | Target sessions over project period | 480         |

Programme List

| ID   | Name                       | Type      | Target Sessions |
|------|----------------------------|-----------|-----------------|
| P001 | After-School Tutoring      | education | 480             |
| P002 | Digital Literacy Labs      | digital   | 300             |
| P003 | Teacher Training Workshops | training  | 120             |
| P004 | Parent Engagement Drives   | community | 200             |

5. Monthly KPIs ( monthly\_kpis.json )

Pre-aggregated monthly metrics for dashboard performance.

| Field              | Type    | Description                       | Example   |
|--------------------|---------|-----------------------------------|-----------|
| month              | String  | Year-month                        | "2025-01" |
| total_sessions     | Integer | Sessions conducted in month       | 65        |
| total_participants | Integer | Total participations (not unique) | 1,050     |

| Field                             | Type        | Description                       | Example   |
|-----------------------------------|-------------|-----------------------------------|-----------|
| <code>total_cost_inr</code>       | Float       | Total spending in month           | 42,500.00 |
| <code>cost_per_participant</code> | Float       | Average cost per participation    | 40.48     |
| <code>cost_per_session</code>     | Float       | Average cost per session          | 653.85    |
| <code>quality_issue_rate</code>   | Float (0-1) | Proportion of records with issues | 0.12      |
| <code>verification_rate</code>    | Float (0-1) | Proportion of verified records    | 0.88      |
| <code>programme_breakdown</code>  | Object      | Per-programme monthly metrics     | See below |

## Programme Breakdown (nested)

| Field                        | Type    | Description                          |
|------------------------------|---------|--------------------------------------|
| <code>sessions</code>        | Integer | Sessions for this programme in month |
| <code>participants</code>    | Integer | Participants for this programme      |
| <code>cost</code>            | Float   | Cost for this programme in month     |
| <code>target_sessions</code> | Integer | Monthly target for comparison        |

## 6. Outcome Summary ( `outcome_summary.json` )

Aggregated outcome metrics with demographic breakdowns.

| Field                                   | Type    | Description                    |
|-----------------------------------------|---------|--------------------------------|
| <code>total_active_beneficiaries</code> | Integer | Currently active beneficiaries |

| Field                   | Type        | Description                            |
|-------------------------|-------------|----------------------------------------|
| total_enrolled          | Integer     | Total ever enrolled                    |
| retention_rate          | Float (0-1) | Active / Enrolled ratio                |
| avg_reading_improvement | Float       | Average reading score gain             |
| avg_math_improvement    | Float       | Average math score gain                |
| avg_digital_improvement | Float       | Average digital score gain             |
| by_gender               | Object      | Metrics broken down by gender          |
| by_category             | Object      | Metrics broken down by social category |
| by_district             | Object      | Metrics broken down by district        |

## 7. Districts ( districts.json )

| Field  | Type   | Description           | Example     |
|--------|--------|-----------------------|-------------|
| name   | String | District name         | "Dharwad"   |
| state  | String | State                 | "Karnataka" |
| region | String | Region classification | "South"     |

### District List

| District | State     | Region |
|----------|-----------|--------|
| Dharwad  | Karnataka | South  |



| District  | State         | Region  |
|-----------|---------------|---------|
| Raichur   | Karnataka     | South   |
| Jhansi    | Uttar Pradesh | North   |
| Fatehpur  | Uttar Pradesh | North   |
| Nandurbar | Maharashtra   | West    |
| Washim    | Maharashtra   | West    |
| Purulia   | West Bengal   | East    |
| Bankura   | West Bengal   | East    |
| Dhamtari  | Chhattisgarh  | Central |
| Korba     | Chhattisgarh  | Central |

## KPI Definitions

| KPI                           | Formula                                                                                                                                        | Level      |
|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| <b>Composite Impact Score</b> | $(\text{reading\_gain} \times 0.3) + (\text{math\_gain} \times 0.3) + (\text{retention\_rate} \times 0.2) + (\text{reach\_factor} \times 0.2)$ | Impact     |
| <b>Retention Rate</b>         | $\text{active\_beneficiaries} / \text{total\_enrolled} \times 100$                                                                             | Outcome    |
| <b>Cost per Beneficiary</b>   | $\text{total\_cost} / \text{active\_beneficiaries}$                                                                                            | Impact     |
| <b>Cost per Session</b>       | $\text{total\_cost} / \text{total\_sessions}$                                                                                                  | Output     |
| <b>Data Quality Score</b>     | $(1 - \text{quality\_issue\_rate}) \times 100$                                                                                                 | Operations |

| KPI               | Formula                                                               | Level      |
|-------------------|-----------------------------------------------------------------------|------------|
| Verification Rate | <code>verified_records / total_records × 100</code>                   | Operations |
| Improvement Rate  | <code>beneficiaries_with_score_gain_&gt;5 / total_active × 100</code> | Outcome    |

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